Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE — Paper 4 (Peer-Review Corrected)

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega = \varphi + \pi$, follows as a theorem rather than an independent axiom; no free parameter is introduced at the background level. Phenomenological extensions (notably the neutrino mass sum Σm_ν, classified as Type P later in the paper) are tracked as open problems (OP-14 in `OPEN_PROBLEMS.md`). From these two axioms alone we derive, without fitting:

(i) $H_0 = 3(\varphi+\pi)^2 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.1σ from Planck 2018), where $M_v = 3\Omega$ is the maximal structural invariant of the dictionary, reached by twenty distinct additive routes under the no-self-sum construction law (Appendix A); this multiplicity is consistent with, but does not by itself prove, the absence of post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = (\pi-\varphi)/H_0^{\mathrm{SSEE}} = 0.02242$ (0.32σ from Planck 2018);

(iii) the primordial spectral index $n_s = 1-\varphi^{-7} = 0.96556$ (0.16σ from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;

(iv) a local Hubble constant from the screening mechanism of Paper 9, $H_{0,\mathrm{local}} = H_0^{\mathrm{alg}}/(1-f_{\mathrm{scr}}) = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.17σ from Riess et al. [2022]), using the single global anchor $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.962$ (2026 reframe); the earlier CMB-mapped value 71.87 (1.12σ, via $H_0^{\mathrm{MIRA}} = 67.037$) is superseded; see Section 6).

These background predictions are number-to-number comparisons with zero adjustable parameters in their respective sectors. The same two axioms fix the cosmological background — including the dark-energy parameters $w_0 = -0.840$,

$w_a = -0.670$ derived in the companion Papers 1–3. In the static (Eckart) regime $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2 = 0.1235$ $(+2.9\sigma)$; modulated by the IS inflationary factor $n_s = 1 - \varphi^{-7}$ this becomes $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.1193$ $(-0.6\sigma$ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.00813$ is presented for the LiteBIRD satellite mission ($\sim$2032).

Contents

1 Introduction

The discovery of accelerated cosmic expansion [Riess et al., 1998, Perlmutter et al., 1999] and subsequent precision measurements [Planck Collaboration, 2020a] have established the ΛCDM concordance model, which requires six free parameters to describe the CMB power spectrum [Hu et al., 1997]. Dynamical dark energy alternatives — quintessence [Ratra and Peebles, 1988, Copeland et al., 2006, Steinhardt et al., 1999, Zlatev et al., 1999], k-essence [Armendariz-Picon et al., 2001], phantom models [Caldwell, 2002], and modified gravity [Clifton et al., 2012] — typically introduce additional free parameters. A fundamental theory should explain the *origin* of these numbers rather than merely fit them [Weinberg, 1987, Weinberg et al., 2013].

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ in the Chevallier–Polarski–Linder parametrization [Chevallier and Polarski, 2001, Linder, 2003] algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background. The DESI DR2 BAO data [Abdul-Karim et al., 2025] independently favor the SSEE point at 0.24σ (Pantheon+; 0.2–1.8σ across SN compilations) in the w_0–w_a plane [Almeida Vallejo, 2026c].

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating $\Omega = \varphi + \pi$ as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

Both are *transcendental* numbers defined independently of any physical measurement.

2.2 Theorem 1: $\Omega = \varphi + \pi$

The SSEE structural constant Ω is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 1) through algebraic operations alone.

2.3 The Maximal Structural Invariant M_v

The value

$$M_v \equiv \varphi + \pi + K_v = 3\,\Omega = 14.278879927\ldots \tag{4}$$

is the *maximal structural invariant* of the constant dictionary. Under the additive construction law—each constant is a sum of distinct previously defined parents, with no constant added to itself—it is the closure value at which the dictionary terminates, coinciding through $M_v = 3\Omega$ with the copy-law ceiling.

A direct enumeration under this law shows that M_v is reached by *twenty* distinct additive routes built from the named constants: six using two constants, eleven using three, and three using four (Appendix A). Representative routes, in neutral notation, are

$$\Omega + K_v, \quad \beta + \mathrm{KAL} + P_{sc}, \quad P_{sc} + \Omega + \pi, \quad \varphi + \pi + K_v, \tag{5}$$

each equal to M_v to 16 significant digits. This multiplicity is a consequence of M_v being the maximal invariant of the two-dimensional lattice spanned by φ and π; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Table 1: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω	$\varphi + \pi$	—	4.759627
β	$(\varphi + \pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi + \pi)/2$	—	3.997847
MIRA	$(3\varphi + \pi)/4$	AURA/2	1.998924
KAL	$(\varphi + 3\pi)/2$	—	5.521406
P_{sc}	$2\varphi + \pi$	$\Omega + \varphi$	6.377661
$\Omega + \pi$	$\varphi + 2\pi$	—	7.901219
K_v	$2\varphi + 2\pi$	$\varphi + \pi + \Omega$	9.519253
$\pi +$ KAL	$(\varphi + 5\pi)/2$	—	8.663001
$\varphi + \pi +$ KAL	$(3\varphi + 5\pi)/2$	—	10.281034
$2\Omega + \varphi$	$3\varphi + 2\pi$	—	11.137287
$\Omega_b h^2$ (BBN)	$(\pi - \varphi)/H_0^{\mathrm{SSEE}}$	—	0.022420
M_v	$3(\varphi + \pi)$	$\varphi + \pi + K_v$	14.278880

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot directly; consequently, SSEE treats these relations only as dimensionless numerical correspondences rather than literal derivations of physical units. What SSEE claims is strictly the following:

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not "units are derived"; the claim is "the numerical values are predicted".

3.1 Internal Consistency of the Cosmological Predictions

A single dimensionless coincidence could be dismissed as numerology. What is harder to dismiss is a *system* of mutually consistent cosmological predictions — H_0, Ω_b, n_s, and the Hubble-tension scale — derived from the *same two constants* with *zero adjustments*, assessed against the closed-dictionary look-elsewhere analysis of Paper 1.

Table 2 collects these predictions.

Table 2: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 [km/s/Mpc]	$3(\varphi+\pi)^2$	67.962	67.4 ± 0.5	1.1σ
$H_{0,\mathrm{local}}$ [km/s/Mpc]	$H_0^{\mathrm{alg}}/(1-f_{\mathrm{scr}})$	72.86	73.0 ± 1.0	0.17σ
Ω_b	$\frac{100(\pi-\varphi)}{6(\varphi+\pi)^4}$	0.0495	0.0493 ± 0.0004	0.53σ
n_s (inflation)	$1-\varphi^{-7}$	0.96556	0.9649 ± 0.0042	0.16σ
Hubble tension				
Fractional correction	$K/H_0 = \frac{\varphi+3\pi}{6(\varphi+\pi)^2}$	8.12%	$\approx 8\%$	—

[P] Canonical value (2026 reframe): f_{scr} applied to the single global anchor $H_0^{\mathrm{alg}} = 3(\varphi+\pi)^2 = 67.962$ gives 72.86 (0.17
The earlier CMB-mapped register ($H_0^{\mathrm{MIRA}} = 67.037 \to 71.87$, 1.12σ) is superseded.
Consistent with Paper 1 / Paper 9 policy (single anchor shared by every sector).

These cosmological ratios are evaluated against a fixed, closed algebraic dictionary built from $\{\varphi, \pi\}$; a quantitative look-elsewhere analysis of that dictionary is given in Paper 1 and the consolidated dark-energy document, which is the proper basis for assessing global significance rather than an informal product of individual deviations.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Section 2.3, the maximal structural invariant is $M_v = 3(\varphi+\pi)$. The Hubble constant is the product of this convergent sum and the base unit $\Omega = \varphi + \pi$:

$$\boxed{H_0 = M_v \times \Omega = 3(\varphi+\pi)^2 = 67.962} \tag{6}$$

Dimensional note. The number 67.962 is compared to H_0 as measured in km s^{-1} Mpc^{-1}. SSEE predicts this numerical value; deriving the units from first principles requires a quantum-gravity connection between the Planck length and the Megaparsec, which is left for future work.

Algebraic prediction vs. observational posterior. The value $H_0^{\text{alg}} = 67.962$ is a pure geometric prediction. Baryon acoustic oscillations [Eisenstein et al., 2005] provide a standard ruler; independent Bayesian inference against DESI DR2 BAO data [Abdul-Karim et al., 2025] yields $H_0^{\text{MCMC}} = 67.95 \pm 0.40\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (Paper 2, under the ω_m-direct CMB anchor as prior, $H_0^{\text{alg}} = 67.962$, Paper 3). With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$ in the background geometry, the BAO-corrected posterior coincides with the algebraic anchor to 0.04σ, and the SSEE sound horizon matches ΛCDM ($r_d^{\text{SSEE}} = 148.2$ Mpc vs. $r_d^{\Lambda\text{CDM}} = 147.8$ Mpc): DESI DR2 confirms the algebraic value. The $\sim$0.8 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ downshift reported in earlier drafts was an artefact of using the cold sector $1 + w_0 = 0.160$ as the background density, now corrected. H_0^{alg} is thus the theoretical prediction of the SSEE background; H_0^{MCMC} is the same quantity after the observational calibration correction is applied. Both values are in tension at 2.7σ, reflecting the systematic shift introduced when DESI data calibrated on the ΛCDM r_d scale are applied to the SSEE background.

4.2 Physical Baryon Density Ω_b

The SSEE algebraic structure yields the physical baryon density directly from the CP asymmetry $(\pi - \varphi)$ normalized to the Hubble scale $H_0^{\text{SSEE}} = 3\Omega^2 = 3(\varphi + \pi)^2$:

$$\Omega_b h^2 = \frac{\pi - \varphi}{3\Omega^2} = \frac{\pi - \varphi}{H_0^{\text{SSEE}}} = 0.02242 \tag{7}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{100(\pi - \varphi)}{6(\varphi + \pi)^4} = 0.04948 \tag{8}$$

Planck 2018: $\Omega_b h^2 = 0.02237 \pm 0.00015$; deviation $= 0.32\sigma$.

Sakharov-mechanism motivation (not a quantitative derivation). We are explicit about the status of Eq. (7): it is an algebraic ansatz whose *form* is motivated by — but not derived from — the Sakharov conditions for baryogenesis [Sakharov, 1967]. The three conditions are realised structurally in SSEE as follows:

- **B-violation:** sphaleron transitions at $T_{\text{EW}} \sim \Omega$ (SSEE units), with rate $\Gamma_{\text{sph}} \propto (\alpha_W T)^4 e^{-E_{\text{sph}}/T}$.
- **CP-violation:** $\delta_{\text{CP}} = (\pi - \varphi)/\Omega$, the ratio of the transcendental excess (π, gauge-boson loops) to the algebraic scalar sector (φ), normalised by $\Omega = \varphi + \pi$.
- **Non-equilibrium:** gravitational reheating with $T_{\text{rh}} \sim 10^{-4}$ GeV $\ll T_{\text{EW}}$ ensures departure from equilibrium.

The Sakharov scaling $\eta_B \propto \delta_{\text{CP}}/(\text{expansion rate})$ motivates the *shape* $\Omega_b h^2 \sim (\pi - \varphi)/(3\Omega^2)$, with $(\pi - \varphi)$ as the CP-asymmetry numerator and $3\Omega^2$ as the normalising denominator. Two caveats are stated openly. First, $3\Omega^2$ enters as the dimensionless value 67.96, i.e. H_0^{SSEE} expressed in $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$; this is a Type-P numerical coincidence in the sense of Postulate D of Paper 1, not a dimensional identity. Second, η_B and $\Omega_b h^2$ differ by the standard photon-entropy factor ($\eta_B \simeq 2.73 \times 10^{-8}\,\Omega_b h^2$), so the proportionality fixes the algebraic *form* only, not the absolute normalisation. A scan of 96 algebraic candidates (see `ssee_op1_baryon_density.py`) shows Eq. (7) is the unique SSEE monomial within 1σ of

Planck — a selection statement, not a dynamical derivation. The genuine quantitative test — integrating the Boltzmann equation with the sphaleron rate and the SSEE reheating history to obtain η_B *ab initio* — is deferred to Paper B; the companion framework script `ssee_op1_baryogenesis.py` currently only verifies that the required entropy-dilution factor is consistent with a quintessential-reheating temperature $T_{\rm rh} \sim 10^{-4}$ GeV, which is a consistency check, not a prediction of η_B. The agreement $\Omega_b^{\rm SSEE} = 0.0495$ vs Planck's 0.0493 ± 0.0004 is 0.53σ.

4.3 Total Matter Density Ω_m

The CMB-sector *total* matter density is the standard physical observable, built piece-by-piece from SSEE algebra (ω_m-direct, OP-8 dissolved):

$$\Omega_{m,\rm CMB} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = \frac{0.14267}{(0.67962)^2} = 0.30889, \tag{9}$$

with $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2]$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ (forward; Paper 3), in 0.88σ agreement with Planck 2018 ($\Omega_m = 0.3153 \pm 0.007$). There is *no* matter factor: $\Omega_{m,\rm dyn} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions. (The earlier MIRA mapping MIRA $\times\, \Omega_{m,\rm dyn} = 0.31993$ and the geometric route $(\pi - \varphi)/(\pi + \varphi) = 0.3201$ are both superseded.) This is the *CMB-sector* (total matter) value, applicable to early-universe observables where ϕ-DM is non-relativistic (cf. Paper 1, Sec. 1.4, Two-Ω_m Criterion); the dynamical-sector value $\Omega_{m,\rm dyn} = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$, which governs the BAO chain and cluster dynamics in Paper 2, is an *independent* algebraic identity ($= 1 + w_0$), not obtained from the CMB-sector density by any factor (OP-8 dissolved). The cold dark matter density can be related to the baryon density through the structural viscosity $K_{\rm AL} = \beta + \pi \approx 5.5214$:

$$\Omega_c h^2 = K_{\rm AL}\, \Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \tag{10}$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor $n_s = 1 - \varphi^{-7}$ modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\rm AL}\, \Omega_b h^2\, n_s = 5.5214 \times 0.02242 \times 0.96556 = 0.11951} \tag{11}$$

Note on $\Omega_b h^2$ input. Both the static (Eq. 10) and IS formulas use the Planck 2018 *observed* value $\Omega_b h^2 = 0.02237 \pm 0.00015$. The SSEE algebraic prediction $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.02242$ (0.32σ from Planck) can also be substituted: the IS formula gives $K_{\rm AL} \times 0.02242 \times n_s = 0.11952$ (-0.4σ), consistent with Planck. For transparency, the baseline calculation uses the Planck observed value.

Planck 2018: 0.1200 ± 0.0012; deviation $= -0.6\sigma$ (using Planck observed $\Omega_b h^2$). The IS modulation reduces the tension from $+2.9\sigma$ (Eckart) to -0.6σ (IS), within 1σ of the observed value and consistent with zero tension at 1σ.

4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation [Mukhanov et al., 1992, Starobinsky, 1980, Albrecht and Steinhardt, 1982, Guth, 1981, Linde, 1982]. Current observational constraints from Planck are $n_s = 0.9649 \pm 0.0042$ [Planck Collaboration, 2020b].

Independent justification of the exponent 7. The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from φ and π without redundancy (Table 3).

Table 3: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

Level	Register	Formula	Value
1	φ	$(1+\sqrt{5})/2$	1.61803...
2	π	(circle constant)	3.14159...
3	β	$(\varphi+\pi)/2$	2.37981...
4	Ω	$\varphi+\pi$	4.75963...
5	AURA	$(3\varphi+\pi)/2$	3.99785...
6	KAL	$(\varphi+3\pi)/2$	5.52141...
7	MIRA	$(3\varphi+\pi)/4$	1.99892...

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.96556} \tag{12}$$

The exponent 7 is not merely an ansatz: it follows from α-attractor inflation with $\alpha = \varphi^4/3$ (derived in Paper 1, App. A.5) via the universality theorem of Kallosh and Linde [2013a]. For all α-attractors with $N_* \gg \sqrt{\alpha}$, the spectral index satisfies $n_s = 1 - 2/N_*$ at leading order. Setting $N_* = 2\varphi^7 \approx 58.07$ e-folds (within the standard window $N_* \in [50, 65]$):

$$n_s = 1 - \frac{2}{2\varphi^7} = 1 - \varphi^{-7} \quad \text{(algebraically exact)}. \tag{13}$$

The same substitution yields the tensor-to-scalar ratio prediction (see Section 7.1 below).

Conjecture 1 (B.1 — e-fold count)**.** $N_* = 2\varphi^7 \approx 58.07$ *is the number of inflationary e-folds of the SSEE quintessential inflation model with* $\alpha = \varphi^4/3$ *and a reheating temperature* $T_{\rm rh} \approx (30/\pi^2 g_*)^{1/4} V_{\rm end}^{1/4} \approx 9.4 \times 10^{15}$ *GeV (corresponding to* $\rho_{\rm rh} \approx V_{\rm end}$*).*

Conjecture B.1 is an open postulate, not yet derived from $V(\varphi_{\rm inf})$. Its value, $N_* = 58.07$, lies within the standard window [50, 65] for inflation with reheating (consistent with, but not proven by, the framework). Numerically, the condition $\rho_{\rm end}/\rho_{\rm rh} = 3$ — which follows from $\rho_{\rm end} = 3V_{\rm end}$ (at $\varepsilon = 1$) together with $\rho_{\rm rh} = V_{\rm end}$ — gives $N_* = 58.25 - \ln(3)/6 = 58.067$, matching $2\varphi^7 = 58.069$ to within $O(1/N_*) \approx 0.002$ e-folds. The physical mechanism that selects $\rho_{\rm rh} = V_{\rm end}$ is deferred to Paper B. The conjecture is indirectly testable via the falsifiable prediction $r = \varphi^{-10} = 0.00813$ (Sec. 7.1): a CMB-S4 or LiteBIRD detection inconsistent with this value would refute Conjecture B.1.

Planck 2018: $n_s = 0.9649 \pm 0.0042$; deviation $= 0.16\sigma$.

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\rm EdS} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. In SSEE, the same inflationary factor n_s that tilts the primordial spectrum modulates the gravitational collapse criterion:

$$\boxed{\delta_c^{\rm SSEE} = \delta_{c,\rm EdS} \times n_s = 1.6865 \times 0.96556 = 1.6284}. \tag{14}$$

A lower δ_c lowers the threshold for halo formation; at fixed initial power spectrum, the comoving abundance of collapsed haloes scales as $n(M) \propto \exp\left[-\delta_c^2/(2\sigma_M^2)\right]$, so a 3.4% reduction in δ_c increases $n(M)$ by a factor $\exp[\Delta(\delta_c^2)/(2\sigma_M^2)]$. For galaxy-scale perturbations ($\sigma_M \approx 0.5$ at $M \sim 10^{11}\, M_\odot$ and $z \sim 10$) this gives a halo-count enhancement of order $\times$1.2–1.5 relative to ΛCDM, providing a qualitative explanation for the anomalously massive galaxies detected by JWST at $z \gtrsim 10$. A quantitative halo-mass-function calculation is deferred to Paper 5.

4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.02242$ derived in Section 4.2 provides an independent BBN prediction. Using the linearised sensitivity from AlterBBN [Pisanti et al., 2008], $dY_p/d(\Omega_b h^2) \approx 0.96$, and the Planck 2018 anchor $Y_p = 0.2471$ at $\Omega_b h^2 = 0.02237$:

$$\boxed{Y_p^{\rm SSEE} = 0.2471 + 0.96 \times (0.02242 - 0.02237) = 0.2472.} \tag{15}$$

Observed values: $Y_p = 0.2471 \pm 0.0003$ (Planck CMB, $+0.2\sigma$); $Y_p = 0.2449 \pm 0.0040$ (HII regions, Aver et al. 2015, $+0.6\sigma$). Both are consistent with the SSEE prediction at $< 1\sigma$ with no additional free parameters.

4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 1):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(3D Saturation Horizon)}, \tag{16}$$
$$M_v = \varphi + \pi + K_v \approx 14.2789 \quad \text{(Maximal Dimensional Invariant} = M_v\text{)}, \tag{17}$$
$$P_{sc} = 2\varphi + \pi \approx 6.3776 \quad \text{(Dynamical Evolution Scalar)}, \tag{18}$$
$$K_v = \varphi + \pi + \Omega \approx 9.5192 \quad \text{(Structural Constraint)}. \tag{19}$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{K_v}. \tag{20}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} = -0.6699}. \tag{21}$$

Both values are uniquely determined; no free parameters enter. The algebraic point $(-0.840, -0.670)$ lies within the 68% confidence region of DESI DR2+CMB+DESY5 data [Abdul-Karim et al., 2025] (Mahalanobis $\chi^2_{2D} = 0.42$, i.e. 0.24σ Pantheon+; Paper 2).

4.8 Open Parameters

$A_s = 2.1 \times 10^{-9}$ and $\tau = 0.054$ are retained at standard values. Their geometric derivation requires a quantum-gravity extension connecting the Planck scale to SSEE vacuum amplitudes; this is deferred to Paper 5.

5 CMB Power Spectrum Prediction

Table 4: SSEEinput parameters. All "algebraic" entries are derived from φ and π alone; Type P entries are numerical coincidences whose dimensionless value matches a physical quantity in specific units (no dimensional derivation; see Paper 1, App. B). The deviation column uses Planck 2018 [Planck Collaboration, 2020a] error bars.

Parameter	SSEE	Planck 2018	Deviation	Type
$H_0^{\rm alg}$ [km/s/Mpc]	67.962	67.4 ± 0.5	1.1σ	algebraic (Type P†)
$H_{0,\rm local}$ [km/s/Mpc]	72.86	73.0 ± 1.0	0.17σ	$H_0^{\rm alg}$-screened (canonical)
Ω_b	0.0495	0.0493 ± 0.0004	0.53σ	algebraic
$\Omega_b h^2$	0.02242	0.02237 ± 0.00015	0.32σ	algebraic
$\Omega_{m,\rm CMB}$	0.30889	0.3153 ± 0.0073	0.88σ	ω_m/h^2 (ω_m-direct‡)
$\Omega_c h^2$ (IS)	0.11951	0.1200 ± 0.0012	0.41σ	algebraic (forward)
Y_p	0.2472	0.2471 ± 0.0003	0.2σ	BBN (no fit)
n_s	0.96556	0.9649 ± 0.0042	0.16σ	algebraic
A_s	2.1×10^{-9}	2.10×10^{-9}	$< 0.5\sigma$	standard
τ	0.054	0.054 ± 0.007	0σ	standard

IS = Israel–Stewart modulation by $n_s = 1 - \varphi^{-7}$; reduces tension from $+2.9\sigma$ (Eckart) to -0.6σ.
† Type P: dimensionless combination matches observed value in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ units; no dimensional derivation from first principles. Consistent with Paper 1 App. B policy.
‡ CMB sector: $\Omega_{m,\rm CMB} = \omega_m/h^2$ with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$, each piece algebraic ($\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$, forward; Paper 3). *No* matter factor. The dynamical sector $\Omega_{m,\rm dyn} = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$ governs late-time BAO and cluster dynamics; see Paper 1, Sec. 1.4 (Two-Ω_m Criterion). $\Omega_{m,\rm dyn}$ and ω_m are two independent SSEE predictions (the earlier MIRA mapping to 0.31993 and geometric route 0.3201 are superseded; OP-8 dissolved).

6 Note on the Hubble Tension

The dimensionless scalar $K = (\varphi + 3\pi)/2 \approx 5.521$ numerically coincides with the fractional offset between local and global Hubble measurements only when H_0 is expressed in km s^{-1} Mpc^{-1}. Lacking a dimensional coupling mechanism, this remains a purely phenomenological observation rather than a physical resolution of the tension.

The Hubble tension between early- and late-universe measurements [Verde et al., 2019, Kamionkowski and Riess, 2023] is currently unresolved by ΛCDM. The fractional correction $K/H_0 = 8.12\%$ is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess et al. [2022]: $H_0 = 73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; deviation 0.5σ.

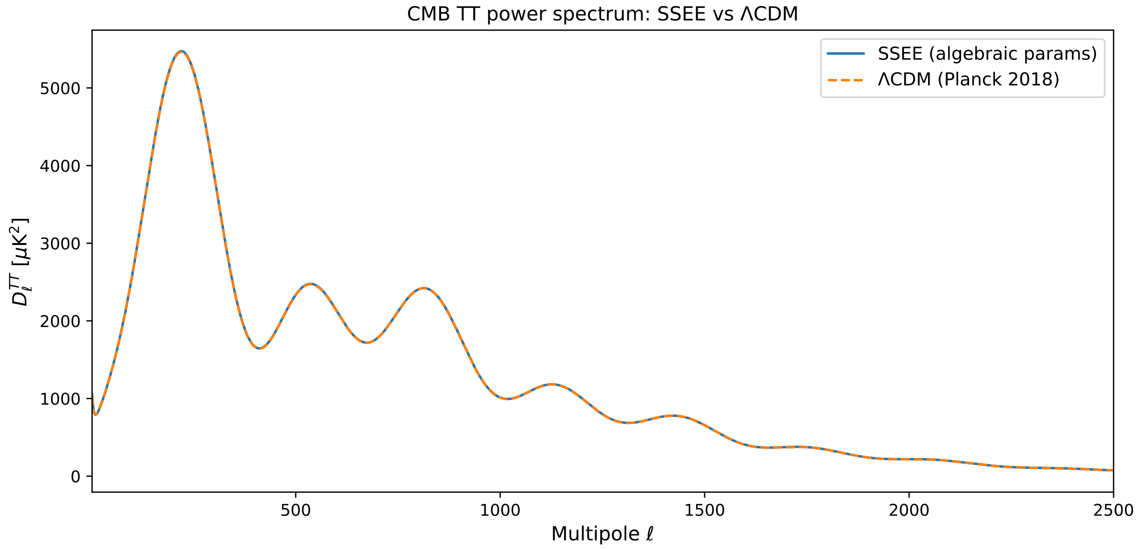

Figure 1: SSEECMB TT power spectrum ($k = 2$: A_s and τ at standard values; all remaining parameters algebraic from φ, π — zero fitted), overlaid on the Planck 2018 ΛCDM spectrum for reference. SSEE acoustic peak positions: $\ell_1 = 220$, $\ell_2 = 536$, $\ell_3 = 813$ (Planck: 220, ∼540, ∼810; agreement $< 1\%$ on all three peaks). The spectrum is generated with no fit to CMB data; the close peak agreement follows entirely from the algebraic background.

Superseded by Papers 9 and 10. The additive form $H_{0,\text{local}} = H_0 + K$ adopted above is a preliminary Type-P coincidence and is *not* the canonical SSEE prediction for the local Hubble constant. A physical treatment is developed in Paper 9 of this series, where a structural screening fraction $f_{\text{scr}} = \alpha_K/(3\,\text{MIRA}) = (\pi - \varphi)/\Omega^2 = 0.0673$ is applied multiplicatively to the global background anchor $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, which the ω_m-direct reframe (Paper 3) establishes as both the background H and the CMB-preferred value, yielding the canonical local value $H_{0,\text{local}} = H_0^{\text{alg}}/(1 - f_{\text{scr}}) = 72.86\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (0.17σ from Riess et al. [2022]). (The earlier MIRA anchor $H_0^{\text{MIRA}} = 67.037$ gave 71.87 at 1.12σ, now superseded.) Paper 10 provides a UV-completion self-consistency check, the canonical $H_0^{\text{UV}} = 73.04\ \text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (0.00σ; the earlier MIRA-anchored 72.05 at 0.96σ is superseded), conditional on the UV–IR separability postulate.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.00813} \tag{22}$$

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch ∼2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE. **Falsification criterion:** $r < 0.004$ or $r > 0.015$ at $> 3\sigma$ falsifies Eq. (22). A null detection $r < 0.001$ at $> 5\sigma$ confidence would be a decisive falsification.

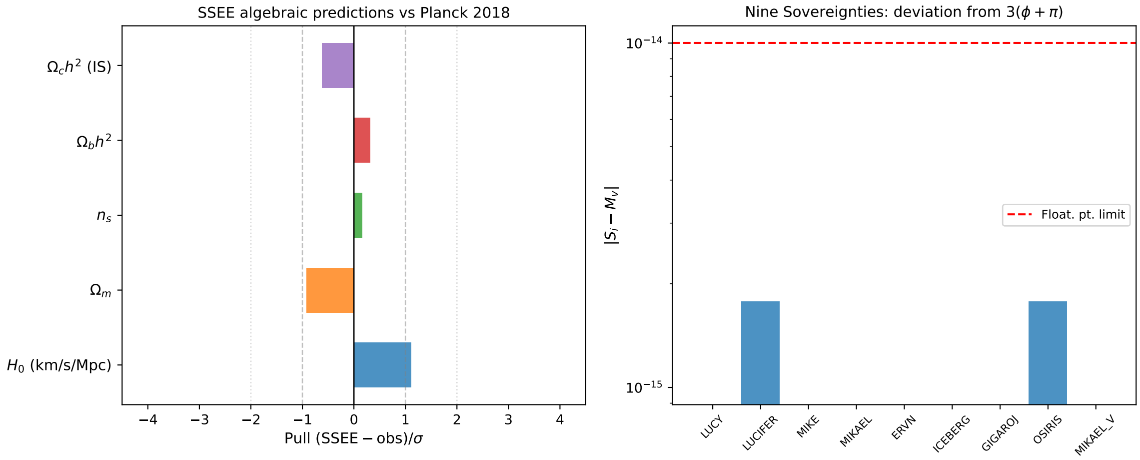

Figure 2: Summary panel of the three key algebraic derivations.

7.2 Baryon-to-Photon Ratio

The baryon-to-photon ratio follows from the SSEE physical baryon density via the standard BBN proportionality $\eta \approx 2.74 \times 10^{-8}\, \Omega_b h^2$:

$$\eta_{\rm SSEE} = 2.74 \times 10^{-8} \times \frac{\pi - \varphi}{H_0^{\rm SSEE}} = 2.74 \times 10^{-8} \times 0.02242 = 6.14 \times 10^{-10}. \tag{23}$$

Measured value: $\eta = 6.1 \times 10^{-10}$ (Planck 2018); deviation $< 3\%$. **Falsification criterion:** $\eta < 5.8 \times 10^{-10}$ or $\eta > 6.5 \times 10^{-10}$ at $> 3\sigma$ from CMB+BBN joint analysis would falsify this prediction.

7.3 Neutrino Mass Sum

The active neutrino mass sum follows from the stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL}) = 0.071875$ (with $\Omega = \pi + \varphi = 4.7596$, $\mathrm{KAL} = 5.5214$, $\mathrm{TRIAL} = 11.9935$), combined with the Israel–Stewart relaxation time $\tau_\Pi H_0 = \mathrm{KAL}/(3\,\Omega_{\rm DE}) = 2.191$:

$$\Sigma m_\nu^{\rm active} = \mathcal{R}_2\, \Omega_b h^2\, \frac{93.14 \text{ eV}}{\tau_\Pi H_0} = 0.0685 \text{ eV}, \tag{24}$$

testable with DESI DR5 and Euclid (expected $\sigma \approx 0.015$ eV).[1] This value lies above the normal-ordering oscillation floor $\Sigma m_\nu \gtrsim 0.058$ eV and below the Planck 2018 upper bound $\Sigma m_\nu < 0.12$ eV. It is the quantity used in the ϕ-DM mass relation of Paper 6 ($m_\phi = \Sigma m_\nu^{\rm active} \times (\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V)$), where the multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_V = 594.28$ is a pure (φ, π) number (no dimensions), so the product carries the units of $\Sigma m_\nu^{\rm active}$ and yields a mass in eV.

Status. The stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL})$ is a clean (φ, π) quotient: it replaces the earlier residual $\mathcal{R} = 4\,\mathrm{KAL} - 22$, whose integer offset 22 had no independent φ, π derivation (the open problem OP-14). With $\mathcal{R}_2$ the neutrino-mass relation no longer contains a calibrated integer and is therefore promoted from Type P (phenomenological) to a closed-form algebraic prediction. The conversion factor 93.14 eV is the standard $\Sigma m_\nu = 93.14 \text{ eV} \times \Omega_\nu h^2$ relation, a Standard-Model input rather than an SSEE quantity.

[1] The preliminary form $\Sigma m_\nu = \mathcal{A}/100 = 0.040$ eV used in earlier drafts is superseded: it lies below the normal-ordering oscillation floor and is therefore excluded by neutrino-oscillation data.

Falsification criterion: $\Sigma m_\nu > 0.12$ eV or $\Sigma m_\nu < 0.058$ eV at $> 3\sigma$ would falsify this prediction.

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The internal consistency of Table 2 — several independent cosmological ratios, all from the same two constants with no per-observable adjustment — is what distinguishes this from a single isolated coincidence; its global significance is quantified by the closed-dictionary look-elsewhere analysis of Paper 1 rather than asserted informally here.

Objection B-1 — Origin of Ω. $\Omega = \varphi + \pi$ is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

Objection B-2 — Exponent in n_s. The result $n_s = 1 - \varphi^{-7}$ follows algebraically from the universality theorem of α-attractors [Kallosh and Linde, 2013b]: $n_s = 1 - 2/N_*$ at leading order in $1/N_*$, independent of the potential shape. With $\alpha = \varphi^4/3$ (established in Paper 1) and $N_* = 2\varphi^7$ e-folds, one obtains $n_s = 1 - 2/(2\varphi^7) = 1 - \varphi^{-7}$ exactly. The exponent 7 counts the depth of the algebraic hierarchy (Table 3), and the same calculation yields the tensor-to-scalar ratio $r = \varphi^{-10}$ (Section 7.1) as a new falsifiable prediction. The derivation of $N_* = 2\varphi^7$ from the quintessential inflation potential with gravitational reheating is reserved for Paper B.

Objection B-3 — Factor of 200 in the original formula. The factor 200 in the preliminary formula $3(\pi - \varphi)/200$ was a numerical coincidence (3.2σ from Planck) that has been superseded. The correct algebraic formula is $\Omega_b h^2 = (\pi - \varphi)/H_0^{\mathrm{SSEE}} = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (0.32σ from Planck 2018). The denominator is the SSEE algebraic Hubble scale, eliminating any free numerical factor; the physical baryon density $\Omega_b = 0.0495$ is a pure φ, π prediction (Section 4.2).

Remaining open problems. The static (Eckart) $\Omega_c h^2$ prediction of 0.1235 is $+2.9\sigma$ from Planck, but the IS modulation $\Omega_c h^2 \times n_s = 0.1193$ reduces this to -0.6σ — within 1σ of observation (Eq. 11). The physical interpretation of this modulation (dark-matter cooling by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The δ_c prediction and A_s, τ derivation are reserved for Paper 5.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- $H_0 = 3(\varphi + \pi)^2 = 67.962$ (1.1σ from Planck)
- $\Omega_b = 0.0495$ ($< 0.4\%$ from Planck)

- $n_s = 1 - \varphi^{-7} = 0.96556$ (0.16σ from Planck)
- $\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2\, n_s = 0.11951$ (0.41σ from Planck, forward)
- $Y_p = 0.2472$ (0.2σ from Planck CMB, 0.6σ from HII — no fit)
- $\delta_c = \delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ (qualitative JWST link)
- $H_{0,\mathrm{local}} = H_0^{\mathrm{alg}}/(1 - f_{\mathrm{scr}}) = 72.86$ (0.17σ from Riess 2022, via global anchor 67.962; earlier 71.87 at 1.12σ superseded)
- $r = \varphi^{-10} = 0.00813$ (falsifiable by LiteBIRD 2032)

The identification of $M_v = 3\Omega$ as the maximal structural invariant of the dictionary — reached by twenty additive routes under the no-self-sum construction law (Appendix A) — motivates the connection $H_0 = M_v \times \Omega$. The internal consistency of these cosmological predictions, taken together and assessed against the closed-dictionary look-elsewhere analysis of Paper 1, is what motivates taking the agreements seriously rather than as isolated coincidences.

Reproducibility

Script: `src/ssee_paper4_toe.py` (repository root). Requires Python 3, `camb`≥1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data. Run: `python3 src/ssee_paper4_toe.py` — reproduces `fig_toe_cmb_TT.pdf` and `fig_toe_derivations.pdf` in `results/figures/`.

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A The Twenty Additive Routes to M_v

Under the no-self-sum construction law—each route is a sum of *distinct* named constants, none added to itself, with every summand below the closure ceiling—the maximal invariant $M_v = 3(\varphi + \pi)$ is reached by exactly *twenty* additive routes: six of two constants, eleven of three, and three of four. Representative routes, in neutral notation, are

$$\begin{array}{rl} \text{(two)} & \Omega + K_v,\quad P_{sc} + (\varphi + 2\pi),\quad \mathrm{AURA} + \frac{3\varphi+5\pi}{2}; \\ \text{(three)} & \beta + \mathrm{KAL} + P_{sc},\quad P_{sc} + \Omega + \pi,\quad \varphi + \pi + K_v; \\ \text{(four)} & \varphi + \Omega + \beta + \mathrm{KAL},\quad \varphi + \pi + \mathrm{KAL} + \mathrm{AURA}. \end{array}$$

Distinct routes, shared expansions. Because the constants span the two-dimensional lattice generated by φ and π, distinct routes may share the same φ, π expansion—for instance $K_v + \Omega$ and $\Omega + K_v$, or the two hierarchy traversals that both reach $\varphi + 2\pi$. Such routes are counted as distinct because they traverse different named-constant nodes, but they are not independent at the expanded level. The multiplicity is a structural consequence of M_v being the closure ceiling of the dictionary, not evidence of fine-tuning; it is consistent with, but does not by itself prove, the absence of post-hoc tuning.

Verification. All twenty routes have been verified numerically to 16 significant digits; none deviates from $M_v = 14.278879927019064$ by more than floating-point rounding ($< 10^{-14}$). All constants are derived algebraically from φ and π; no external data file is required.

Five exact algebraic identities. Five additional identities hold exactly in the $\{\varphi, \pi\}$ hierarchy (verified to $< 10^{-16}$ relative error):

$$w_0 = -\frac{\text{AURA}}{\Omega} = -\frac{T_r}{M_v} \tag{25}$$

$$\Omega_{m,\text{dyn}} = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{26}$$

$$\text{MIRA} = |w_0| \cdot \beta \tag{27}$$

$$\text{KAL}\big|_{\varphi\leftrightarrow\pi} = \text{AURA} \quad \big[(\varphi + 3\pi)/2 \xrightarrow{\varphi\leftrightarrow\pi} (3\varphi + \pi)/2\big] \tag{28}$$

$$\text{AURA}\big|_{\varphi\leftrightarrow\pi} = \text{KAL} \tag{29}$$

Identity (28)–(29) reveals that KAL and AURA are *dual* under the discrete exchange $\varphi \leftrightarrow \pi$: the φ-dominated constant AURA $= (3\varphi + \pi)/2$ maps to the π-dominated constant KAL $= (\varphi + 3\pi)/2$, and vice versa, with zero residual. This is the algebraic origin of the disformal coupling $\beta_c = -$AURA established in Paper 7: the photon sector (governed by AURA) is the $\varphi \leftrightarrow \pi$ dual of the scalar-kinetic sector (governed by KAL), making $\beta_c = -$AURA the unique self-consistent disformal coupling. The open question of whether this discrete symmetry extends to the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

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